

SAM: Simplified Analytical Model of Dike Pathways in Three Dimensions -

Change log - August 2026

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1 Introduction

The new version of SAM (Simplified Analytical Model), a MatLab software performing fast and flexible simulations of three-dimensional (3D) dike pathways in a homogeneous linearly-elastic medium, includes some improvements in the way the model handles dyke propagation within volcanic edifices or topographic reliefs, as well as more options to fine-tune the starting parameters of dyke back-tracking. Lines refer to the current version of the scripts unless otherwise specified. The user is referred to the amended version of ‘SAM Instruction Manual’ and Mantiloni et al. (2023) for further details.

In the following, we list all [changes](#) in the individual scripts with respect to the [original version](#) released in 2023.

1.1 Contact details

Contact details and affiliations were updated in all scripts’ headers. Since work e-mail addresses are liable to change every few years, a personal e-mail address has also been provided.

1.2 SAM

- **Line 76:** change to `if isscalar(TopoInterp) && ~isstruct(TopoInterp)` from `if isscalar(TopoInterp)` to ensure the inputs describing the topography are handled cor-

rectly. In the previous version, providing ‘TopoInterp’ as a 1x1 `struct` would have forced SAM to assume a flat topography at $z = 0$ regardless of the input.

- **Lines 79-83:** field checks for ‘TopoInterp’ change from ‘Xsurf’, ‘Ysurf’, ‘Zsurf’ to ‘Points’ and ‘Elevation’. This reflects the change in how the topography is acquired and described within SAM. The error message was also updated to account for the changes.
- **Lines 85-86:** the ‘Xsurf’, ‘Ysurf’, and ‘Zsurf’ variables are now assigned from ‘TopoInterp.Points(:,1)’, ‘TopoInterp.Points(:,2)’, and ‘TopoInterp.Elevation’ fields, respectively.
- **Lines 87-96:** options for the interpolation and extrapolation methods of the topography interpolation (i.e., how the script assigns an elevation to coordinates which fall inside and outside the input grid describing the topography, respectively) are now assigned from dedicated fields of ‘TopoInterp’. The new fields are ‘TopoInterp.Method’ for interpolation and ‘TopoInterp.ExtrapolationMethod’ for extrapolation. If these fields are not provided by the user, they will be assigned as ‘linear’ by default.
- **Line 97:** ‘Method’ and ‘ExtrapolationMethod’ are added to the arguments of the ‘CreateTopoInterp()’ function.
- **Line 591:** within the ‘CreateTopoInterp()’ function, the interpolation and extrapolation methods are assigned as ‘linear’ by default when the topography is a simple flat surface.
- **Lines 598-599:** still within the ‘CreateTopoInterp()’ function, ‘Method’ and ‘ExtrapolationMethod’ are now retrieved from ‘varargin’.
- **Line 603:** ‘Method’ and ‘ExtrapolationMethod’ are now provided to the ‘scatteredInterpolant’ function.

1.3 SAM_Forward_Interp

- **Line 299:** ‘&& F(3) > -2*MDT’ changed to ‘&& F(3) > -3*MDT’. The aim of this condition, which is only checked when the option ‘ToFreeSurface’ is set to 1 (see the Instruction Manual), is to prevent the dyke pathway from being projected indefinitely downward through the crust if the direction of propagation in the last step of the simulation points downward. In such a case, the When simulating dyke pathways within volcanic edifices, however, dykes might stop well before the Minimum Distance Threshold (MDT, i.e., the minimum distance from the free surface that dykes are allowed to reach, useful when working with Boundary Element numerical models for the crustal stress) is reached. We now set a larger threshold so that the dyke projection to the free surface does not stop immediately when that is the case.

1.4 SAM_Backtrack_Interp

- **Lines 252-257:** a safety check has been added so that when the script accesses the backtracked starting point of the dyke (‘BSP = dike_path_tip(end,:)’), it does not return an error if the ‘BSP’ variable is empty. In that case, the script now returns a warning message and sets ‘BSP’ to [].

1.5 Example1

- **Lines 39-43:** it is now possible to assign two coefficients, '**SH**' and '**Sh**', associated with the most and least compressive horizontal stress component in the crust, to allow for non-isotropic background lithostatic stress. More details are now provided in the 'HalfSpaceNormalForce.m' function.
- **Line 68:** the suggested magma density has been changed to a more commonly-employed value of '**Pm = 2300;**'.
- **Line 76:** changed to '**StressModel.Stressfun_Par = P,Pr,nu,Xloc,Yloc,SH,Sh;**' to account for the new inputs '**SH**' and '**Sh**'.

1.6 HalfSpaceNormalForce

- **Lines 1-2:** the two stress coefficients explained above, '**SH**' and '**Sh**', are added to the input arguments.
- **Header section, lines 20-25:** the description of '**SH**' and '**Sh**' is now included.
- **Line 76:** the horizontal components of the background crustal stress (i.e., the stress before the point-force loads are applied to the surface) are now multiplied by the coefficients '**SH**' and '**Sh**'.

1.7 Example2

- **Header section, lines 40-41:** the variable '**Topo_Interp**' is now defined explicitly in the script rather than imported from 'Example2_Data.mat'. '**Topo_Interp**' is now explained in the header section.
- **Lines 87-89:** '**Topo_Interp**' is now defined as a structure with two fields, '**Topo_Interp.Points**' (corresponding to the x-y coordinates of the mesh nodes describing the topography) and '**Topo_Interp.Elevation**' (corresponding to the elevation of each point above or below the datum level, $z = 0$).
- **Lines 93-94:** a coefficient, '**ElevationScale**', can now be assigned to handle the vertical exaggeration of the mesh plot, so that low-elevation topography can be better visualised.
- **Lines 95-96:** the fourth input argument of `trisurf()` is changed from '**Points(:,4)*5**' to '**Points(:,4)*ElevationScale**'.
- **Lines 178-179:** the fourth input argument of `trisurf()` is changed from '**Points(:,4)*5**' to '**Points(:,4)*ElevationScale**'.

1.8 Example2_Data.mat

The data file for `Example2.m` has been updated, removing the variable '**Topo_Interp**', which was previously defined as a `scatteredInterpolant` variable, and now explicitly defined in `Example2.m`.

2 Acknowledgements

We thank Paul Lundgren for the constructive feedback on the scripts and helping us spot the bugs fixed in the new version.

3 References

Mantiloni, L., Rivalta, E., & Davis, T. (2023). Mechanical modeling of pre-eruptive magma propagation scenarios at calderas. *Journal of Geophysical Research - Solid Earth*.